

Critical Analysis of Mobility and Safety Challenges in Contemporary Space Suit Design

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Abstract

Gas-pressurized spacesuits have effectively been the life-support system throughout the expansion of human space exploration. In conjunction with advanced technology and the evolution of space exploration mission architecture, the design of the gas-pressurized space suit has also been updated. As space technology and mission strategies advance, the design of these suits has evolved, yet current models remain heavy, rigid, and cumbersome, posing challenges for upcoming lunar and Martian missions. These missions demand extensive body movements during extravehicular activities (EVAs), and the current suit designs are not effective in supporting the astronaut's functionality and capability. Lunar and Martian EVAs require a spacesuit to offer flexibility primarily in lower-body movements. A flexible spacesuit made of soft material accommodating a less complex support system will make the suit light, storable, and easy to handle. A less complex life support system operating at higher atmospheric pressure than the current spacesuits could shorten pre-breathe time, which could reduce the pre-EVA preparation time. This paper explores the necessary advancements to create a spacesuit suitable for future planetary missions, spacewalks, and launch-and-re-entry operations.

Keywords: Mobility, Flexibility, Usability, Protection, Spacesuit

Acronyms/Abbreviations

IVA: Intravehicular Activity
EVA: Extravehicular Activity
IEVA: Intra/extravehicular Activity
LES: Launch Entry Spacesuit
ACES: Advance Crew Escape Spacesuit
EMU: Extravehicular Mobility Unit
MCP: Mechanical counter-pressure
CFM: Cubic Feet per Minute
psi: Pounds per Square Inch
psia: Pounds per Square Inch Absolute
CDMA: Code-Division Multiple Access
NASA: National Aeronautics and Space Administration
JAXA: Japanese Aerospace Exploration Agency
GCR: Galactic Cosmic Rays
LEO: Low Earth Orbit

environment. It is a crucial part of the space mission, having evolved significantly over half a century. While developing technology has been contributing to improved usability of spacesuits, the current design is still inadequate for future lunar and Martian missions requiring higher-mobility taskings. Future missions to the Moon and Mars, driven by advanced technologies and a deeper understanding of planetary science, will require astronauts to perform tasks requiring greater mobility. The current spacesuit designs possess limitations to support the high-movement activities. This paper studies the current spacesuits, their limitations, and the scope of improvement to support future space exploration missions.

1. Introduction

The spacesuit functions as a miniature spacecraft that protects astronauts from the inhospitable space

2. History of Spacesuit Design

The journey of spacesuit development began back in the 1930s when pressurized suits were being designed for military applications and stratospheric missions [1]. In

the beginning, the aim was to meet the high-altitude mission requirements for intravehicular activities (IVA). Later, spacesuit designs were separated by functionality such as IVA spacesuits, EVA spacesuits, IEAV spacesuits, LES (Launch Entry Spacesuit), ACES (Advance Crew Escape Spacesuit), Shuttle Space Suit, etc. The USA, Russia, and China are leading designers of spacesuits, however, Japanese spacesuits may soon enter the market, adding to the global diversity of innovations.

2.1 American Spacesuit:

In the USA, the spacesuit development series started with the Mercury Spacesuit, which was designed to support limited intravehicular activities to support a manned Earth orbital mission. The design legacy started with the Mercury project and continued through various iterations including the Gemini Spacesuit, Apollo Spacesuit, Apollo-Soyuz Test Project Spacesuit, Skylab Spacesuit, Space Shuttle Spacesuit, Space Station Spacesuit, and commercialized suits from SpaceX, Boeing, Blue Origin, etc.



Fig. 1. NASA EMU [2]

Each successive series has seen enhancements aimed at improving astronaut protection during space exploration missions for a short to long period. In this

R&D series, the following characteristics have been noted in the American Spacesuit designs.

- Made of 14 layers of flexible material [3]
- Outer layers are made of hard materials, especially the torso part, which is made of glass fiber to reinforce the fabric
- Joints in the torso section such as at the shoulder and lower back have aluminum bearings
- Weighs around 120kg [3]
- Operates at 0.3 atmospheric pressure [3]

2.2 Russian Spacesuit:

The Russian spacesuit technology during the early era of space exploration from the 1930s to 1960s closely mirrored the American designs largely due to the shared German technological heritage. However, following the 1960s, the Soviet Union put efforts into developing different types of spacesuits per the mission requirements such as EVA, IVA, Lunar spacesuit, etc. Early full-pressure suit designs lacked flexible joints, making the movements of arms and legs difficult for astronauts when the suit was pressurized, rendering them unsuitable for space missions [4]. During the post-World War II era, several technical innovations were witnessed by the defense industry; some of them such as pressurized cockpits in jets, ejection seats, and life support systems (oxygen supply) resulted in the development of a full pressurized suit for high altitude missions. The full pressurized suit was stated to contain:

- A ventilation system with a 1 bar pressure to support airflow rate varying from 50 to 150 CFM/min for ten days in open loop situations and for 14 hours in closed loop situations [4].
- KP-50 oxygen unit to provide oxygen supply during the mission, before seat ejection, and after seat ejection [4].
- A bio-temperature regulation system when the cabin temperature goes high up to 40°C [4].

Prior to the Vostok project, the Russian space program had less than five operational EVA/IVA suits; the Vostok project was designed to test no less than 48 spacesuit designs. The project started with the first design containing [4]:

- Mass: 23kg
- Nominal operation period after manufacturing the suit: 6 months

- Two-layer enclosure with a separate load-bearing and pressure-tight enclosure
- Dual bladder and two operating pressure modes (400 and 270 hPa)
- Helmet with a dual visor, a device for automatic closure of the visor
- Removable gloves and cuffs
- Internal thermal protection suit with a ventilation system
- Boots adopted for parachute landing
- Rescue float support collar with a CO₂ inflation system
- Group suit inlet interface unit
- Headset
- Emergency Radio Set
- Survival kit useful if landed in the wilderness

The current Russian spacesuit mirrors the American spacesuit design in some areas, but it operates at 0.4 atmospheric pressure compared to the American spacesuit which operates at 0.3 atmospheric pressure [3].



Fig. 2. Orlan-D [5]



Fig. 3. Orlan-MK [6]

2.3 Chinese Spacesuit:

The Chinese Feitian spacesuit shares similarities with the Russian Orlan spacesuit. It was used for the first time for an 18-minute EVA in 2008. However, public information about the Feitian spacesuit is very limited.

Characteristics of a Feitian spacesuit [7]:

- Weight: 120kg
- Accommodates occupant of height: 1.6 m to 1.8 m
- Nominal pressure: 5.7 psi
- Life Supported System located inside the suit
- Active thermal control system
- The visor is wider than the Orion one
- The helmet is assumed to be built onto the torso
- Two umbilicals at 4m length and 8m length
- EVA control panel is the same as Orlan
- Digital signals-based technology for data collection and transmission
- CDMA-based communication system
- Softer joints than Orlan - Knee and Elbow
 - Flexible limb joints
- Surface material:

- Advanced synthetic fabric
- Fire and radiation resistance
 - Bears temperature up to 200°C
- Wear and tear resistance
 - Protection from floating micro-meteorites
- Pre-breath time: 100 minutes



Fig. 4. Chinese Feitian Spacesuit [7]



Fig. 5. Chinese Feitian Spacesuit [8]

3. Limitations of the existing spacesuit design

The current spacesuit design has evolved through various experiments but still holds some limitations that affect astronauts' comfort, mobility, and ultimately the effectiveness of the mission. The current spacesuit design is not fully equipped to support future long-duration space exploration missions, especially on the lunar and Martian surfaces. This section identifies the limitations of the current spacesuit design in the most impactful areas.

3.1 Pressure:

The duration of the pre-breathe protocol for a spacesuit is determined by the air pressure at which the suit operates. The greater the disparity between the spacecraft's internal pressure and the pressure maintained within the spacesuit, the longer the pre-breathe time needed. Pre-breathe protocol is a crucial protocol for preventing decompression sickness in astronauts during spacewalks by extracting nitrogen from the body. The current American spacesuit operates at 0.3 atmospheric pressure while the Russian spacesuit at 0.4 atmospheric [3], which requires at least 60 minutes of the pre-breathe protocol before astronauts go on spacewalk [9]. The pre-breathe time is protocol-based, therefore it could go from an hour to a few hours before the spacewalk, which increases the pre-EVA preparation time.

3.2 Torque Mobility:

The mobility of spacesuits, particularly in terms of torque, is a significant challenge. This affects how astronauts can maneuver and perform tasks in space. The rigidity of joints in spacesuits is one of the primary factors that limit torque mobility.

3.2.1 Rigid Joints:

Current spacesuits, including the Extravehicular Mobility Unit (EMU), Advanced Crew Escape Suit (ACES), and Mark III, face significant challenges in joint mobility due to rigid components [10]. While rigidity is necessary for maintaining structural integrity and pressure but severely restricts the range of motion, increasing the force required to move the joints, known as joint torque. The stiffness and limited mobility can lead to increased fatigue for astronauts during extravehicular activities (EVAs). Studies show that joint torque is a critical factor affecting the efficiency of EVA tasks, and reducing this torque can significantly improve astronaut performance [11].

3.3 Usability:

Spacesuits function as small, self-contained spacecraft, integrating various systems to protect astronauts from the harsh space environment, which makes the design complex and a bit challenging to use. This complexity compromises bulkiness, mass, size, wear and tear, cleanliness, maintenance, etc [12].

3.3.1 Bulky Design:

Most modern and past spacesuits utilize a gas-pressurized system to maintain optimal internal pressure for the wearer [13]. However, due to the numerous layers and installations, this design often results in bulky, large, and heavy suits. Even experienced subjects frequently display inconsistency in their movements, with over 50% of analyzed motions showing statistical inconsistency when using the MARK III spacesuit [14].

3.3.2 Weight and Size:

MARK III spacesuits have multiple layers of fabric design, intended for protection against extreme pressures and temperatures, which contributes to their substantial weight. For example, the minimum mass of the MARK III was 36 kg (with the incorporation of lighter materials), but it can weigh up to 74 kg or potentially more [15]. Considering that the gravitational acceleration of Mars is 3.72 m/s² compared to 1.62 m/s² of the moon, reducing the weight of spacesuits is crucial for the feasibility of Mars and deep space missions.

A more modern spacesuit like the Z-2 spacesuit of NASA is expected to have a mass of 65 kg, weighing lighter than conventional MARK III spacesuits but still mass-intensive [16].

3.3.3 Transport/Storage:

The substantial weight and size of the spacesuit add constraints to transportation, as volume and mass are already limited in mission planning and execution. Spacesuits require specialized packing and handling to withstand the stresses of launch and space travel, including vibrations and pressure changes. In space, efficient storage solutions are crucial as suits must be stored in compact, accessible compartments without compromising their condition. The design must also facilitate easy donning and doffing in microgravity, requiring innovations to streamline suit deployment and minimize preparation time [17].

3.3.4 Wear and Tear:

Although lethal accidents associated with spacesuits are rare, a critical incident occurred in 2013 when an astronaut experienced water buildup in his helmet during an EVA due to contamination in the water/gas separation filter of the EMU [18].

Gas-pressurized spacesuits are also vulnerable to punctures, which, although less common due to the

multilayered structure, could be catastrophic if they occur [19].

3.3.5 Cleaning:

Dust removal was proven vital based on previous lunar surface missions [20], [21]. Astronauts have reported abrasion of the outer layer used for energy absorption, blurred vision from helmet dust, false instrument readings, seal failures, clogged mechanisms, and inhalation (Overall nine categories of dust problems) [20]. Considering the presence of wind on Mars, improving the dust removal mechanism may be even more necessary to ensure the success and safety of missions [22].

3.4 Biocompatibility:

Beyond mobility, spacesuit design must prioritize the wearer's comfort and health, especially during extended missions. This necessitates selecting materials that prevent skin irritation, regulate moisture and temperature, and are suitable for prolonged wear. Historically, spacesuits were made with synthetic fibers like nylon and polyester, which often led to skin irritation from friction and poor moisture management. Soft rubber components, such as gloves, boots, linings, and gaskets, also showed significant wear and degradation over time. Early testing environments, such as swimming pools, may have skewed material performance results, leading to suboptimal material choices, including natural rubbers and aluminum foils due to their availability.

Today's spacesuits, however, leverage advanced materials like medical-grade silicones and low-friction coatings, designed to enhance durability and wearer comfort while addressing the challenges of long-term space exploration.

4. The scope of improvement

4.1 Pressure:

The increased air pressure in the spacesuit will reduce the air pressure difference between the spacesuit and the spacecraft, which will reduce the pre-breathe protocol time. Though the higher air pressure in the spacesuit will reduce the pre-space mission execution preparation time, the air pressure cannot be increased in spacesuits above a certain limit as it makes the suit swell in the vacuum and makes it rigid, which makes the movement difficult.

Future missions will require more than one spacesuit design - one that the spacesuit can be used for spacewalk and one that will be used for lunar and martian missions. The air pressure difference between the spacecraft cabin and the habitats on the lunar and Martian land will drive this requirement.

The air pressure in the spacesuit is not only determined by the air pressure in the spacecraft cabin/space habitats but it is also limited by the spacesuit design. A pre-breathe protocol was performed at 10.2 psia for several EVAs that were performed on the Space Shuttle [23]. Reasonably higher air pressure in the spacesuit can be achieved by using a material that can hold the higher air pressure by not making the suit rigid, improving the life support system to maintain higher air pressure in the suit, and storing enough oxygen to support the increased air pressure and a structure of a suit that can incorporate the improved life support system.

4.2 Torque Mobility:

Enhancing the mobility of spacesuits, particularly in terms of torque, involves using advanced materials and innovative joint designs. Recent advancements include the development of flexible joints and soft robotics, which significantly reduce the force required for movement, thereby decreasing astronaut fatigue during extravehicular activities (EVAs). Additionally, incorporating materials like shape-memory alloys and high-strength polymers can further improve joint flexibility and overall suit performance, enabling astronauts to move more freely and efficiently in the challenging environments of space.

4.2.1 Flexible Joints :

Incorporating flexible materials and advanced joint designs can enhance torque mobility. Mechanical counter-pressure (MCP) suits, which apply uniform pressure directly to the skin, show promise in reducing joint torque and improving overall mobility. These suits are designed to mimic the flexibility of human joints, allowing for a greater range of motion and reduced fatigue during EVAs [24], [25].

To address this difficulty, some studies conducted in the past include the following designs:

- **Origami-Inspired Joint Designs:** These joints are made from flexible “origami” parts supported by artificial muscles, allowing for high mobility and functionality under relatively

high operating pressures. This design significantly improves the flexibility and range of motion, making the suits more adaptable to the dynamic movements required during planetary exploration.

- **Soft Exoskeleton Knee Prototypes:** Developed to enhance mobility in advanced space suits, these prototypes also use pneumatic actuators to provide additional strength and flexibility to astronauts’ movements, reducing the physical strain and improving overall mobility during EVAs.

4.3 Usability:

Numerous efforts have been made to resolve some of the previously mentioned limitations. One example of this is NASA’s Z series spacesuit, which, despite delays in 2014, aims to enhance usability through multiple improvements [26].

The current Z series spacesuits use a traditional gas-pressurized system, but research into mechanical counter pressure (MCP) holds promise.

4.3.1 Bulky Design:

There are two main ways to mitigate the limitations related to bulkiness, weight, size, durability, and storage. One potential direction is to reduce the number of layers in spacesuits by utilizing multifunctional fabrics. For example, Vectran, a durable variant of Kevlar, is highly resistant to tears, impacts, and extreme temperatures [27]. There is a need for further testing with Vectran, but it shows potential as a multifunctional fabric that can help mitigate some of the limitations. Kevlar is currently being studied by NASA [28].

Another direction is to use the MCP system for spacesuits. MCP spacesuits apply pressure directly to the wearer using tight-fitting materials instead of using gas, offering multiple advantages over traditional gas-pressurized spacesuits [21]. This can reduce the number of necessary layers, resulting in more compact, lighter, and mobile spacesuits.

4.3.2 Weight and size:

The weight and size of spacesuits have been a critical consideration since their inception, given the physical constraints they impose on astronauts during extravehicular activities (EVAs). Traditional gas-pressurized suits are particularly bulky, as they must maintain a stable internal environment, leading to

multiple layers of protective materials that add to their overall mass. According to recent studies, materials like Vectran, a durable variant of Kevlar, have shown promise in reducing the weight of spacesuits while maintaining structural integrity and protection against extreme space conditions [29].

The advancements in material science, such as those discussed by Patel et al., suggest the possibility of utilizing lighter, stronger materials that could significantly reduce the overall weight of the suit without compromising safety. These developments could lead to more compact and efficient spacesuits, which would not only enhance astronaut mobility but also reduce the physical strain experienced during long-duration EVAs. Additionally, the trend toward modular design allows for better size customization, further reducing unnecessary bulk and optimizing the fit for individual astronauts.

4.3.3 Transport/Storage:

The transportation and storage of spacesuits present unique challenges, especially in the context of long-duration missions where space and weight are at a premium. Traditional spacesuits require significant storage space due to their rigid structure and multiple layers of material, which can complicate logistics on missions to the Moon, Mars, or deep space. According to NASA's Office of Inspector General, maintaining and transporting these suits is becoming increasingly difficult as they continue to age and exceed their intended lifespan.

In addition to weight considerations, efforts to streamline the storage of spacesuits include the development of more flexible materials and the potential use of mechanical counterpressure (MCP) technology, which could allow for suits to be more easily compressed and stored. This would not only save space but also reduce the overall payload weight, a critical factor in mission planning and execution. The continued focus on reducing the storage footprint of spacesuits will be essential for the success of future exploration missions [17].

4.3.4 Wear and Tear:

MCP can also enhance the safety of spacesuits. A tear in a traditional gas-pressurized suit would cause fatal damage due to the pressure difference between the interior and exterior. However, the elasticity of the MCP

suits would help minimize and localize the damage from a tear.

4.3.5 Cleaning:

The current planetary dust cleaning or mitigation system for spacesuits should aim to resolve previous sensitivities discovered during the Apollo missions.

The current xEMU demo presented in 2018 by NASA includes an environmental protection garment (EPG) that protects the spacesuit from dust [30]. EPG development, which is projected to end in Aug 2024, will provide active and passive dust removal while being 25% lighter than the previous [31]. Other strategies may include dust-resistant coatings, durable microfiber wipes, or even electrodynamic dust shielding (EDS) technology.

4.4 Biocompatibility:

In the early stages of spacesuit development, managing moisture and hygiene posed significant challenges. Traditional synthetic fabrics, though durable, were unable to effectively wick away moisture, often causing discomfort and skin irritation due to sweat buildup. Additionally, early designs did not consider the growth of microbes, which posed risks during extended missions. Over time, advancements in fabric technology introduced moisture-wicking materials that moved sweat away from the skin, thus improving comfort. Antimicrobial treatments were also incorporated to mitigate the growth of bacteria, ensuring better hygiene for astronauts on longer missions.

Temperature regulation, another crucial aspect, was originally managed by liquid cooling and ventilation garments, which circulated water through tubes embedded in the suit to maintain body temperature. These garments remain a key element of modern spacesuits, but future designs are set to enhance this system through phase-change materials (PCMs). PCMs absorb and release heat as they transition between solid and liquid states, providing more efficient thermal management. This helps reduce weight and complexity while keeping astronauts comfortable in extreme temperatures.

Looking to the future, smart textiles are being explored as a transformative step in spacesuit design. These textiles can react to environmental changes or physiological cues, such as temperature or moisture levels, automatically adjusting to optimize comfort. For example, fabrics embedded with sensors and

micro-heating elements could regulate warmth based on the wearer's body heat or external conditions, creating a dynamic, responsive suit that adapts in real time. Smart textiles may also include self-repair capabilities, increasing durability during long-term missions in space.

Modular suit designs are another area of innovation that aims to improve the fit for diverse body types. In the past, spacesuits were custom-tailored to individual astronauts, limiting flexibility and requiring extensive pre-flight preparation. Now, research is focused on modular components that can be easily adjusted or swapped to fit different body shapes and sizes, improving accessibility and wearability for a broader range of astronauts.

5. Designs under test

5.1 Japan Spacesuit:

JAXA has always made critical contributions to the space industry through advanced technology. Lately, JAXA has started working on designing a spacesuit to support future space exploration missions by addressing the limitations of the current spacesuits. JAXA is focusing on a major material change; per the interview of Yoshio Wada, an engineer at the Human Space Systems and Utilization Mission Directorate, JAXA, the Japanese spacesuit will have the following characteristics [3]:

- The spacesuit will be made of only fabric - foldable, compact, easy to store, protection against shock and vibration during the launch and re-entry, etc.
 - The same spacesuit can be used for launch and re-entry and EVAs.
 - Garment for liquid cooling and ventilation
 - Garment withstand the air pressure in the suit
 - Soft fabric to support extensive movements
- The spacesuit will operate at 0.58 atmospheric pressure - reducing pre-breathe time
- The spacesuit will weigh about 90kg - lightweight

Per the interview, it is understood that the prototypes of pressure resistance and liquid cooling and ventilation garments are ready. Further information is not available to the public, therefore there are limitations to learning the fabric.

5.2 Axiom Extravehicular Mobility Unit (AxEMU):

Axiom spacesuit, which is known as the Axiom Extravehicular Mobility Unit (AxEMU) is designed by Axiom private corporation, which is collaborating with NASA to support NASA's future space exploration missions, especially lunar missions. Per the announcement made in 2023, the AxEMU will be used on the Artemis III mission in 2026 [32]. Further research is going to make the AxEMU suitable for the ISS missions.

The material and design of this suit are proprietary, therefore specifics of the design and material are not available publicly. Based on reliable article sources, it is understood that this spacesuit is composed of modular design to support LEO-based missions and beyond LEO-based missions. The adaptability to the LEO space environment and beyond the LEO space environment is achieved through the material(s) used for radiation and thermal protection. The corporation claims the suit's advanced capabilities support high mobility activities. The AxEMU design supports the sustainability goal by enhancing the reusability of the suit for multiple missions.

Figure 6 demonstrates the Axiom Extravehicular Mobility Unit Spacesuit launched in the market. Considering the thermal regulation requirements of the spacesuit, the exterior material of the suit will use the traditional white color in the proposed material of AxEMU instead of a black material. The material used in this layer is proprietary, therefore further information on this material is not available publicly.



Fig. 6. AxEMU [33]

5.3 Shrink Wrapping Spacesuit:

A shrink-wrapping spacesuit is also referred to as a second skin suit. Various institutes are doing research on this concept, but MIT researchers are getting closer to launching this second-skin suit. As the name suggests, this concept provides a skinnier suit that is light in weight, flexible, and comparatively easy to wear. Several materials were studied from dielectric elastomers to shape-memory polymers to achieve flexibility in the suit. Based on the material analysis, the MIT research group led by Prof. Dava Newman and her postdoc candidate Bradley Holschuh finalized the Nickel-Titanium shape memory alloy. The suit design proposed by this team contains very thin straight fibers of nickel-titanium shape memory alloy, which transform into coils that help the suit to skin fit the astronaut's body. These coils require high pressure to contract the suit to make it fit the astronaut's body. This pressure is applied by running current through these coils, which requires heavy battery packs. The suit's dependency on the heavy battery packs makes it unreasonable and unsuitable for space missions. Prof. Newman and Holschuh are exploring new techniques to latch the coils in place.

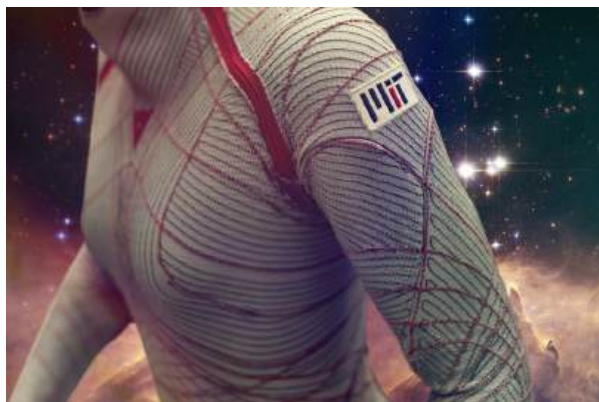


Fig 7. Shrink-wrapping Spacesuit [34]

6. Recommendations

6.1 Material:

The lunar, Martian and other potential deep space exploration require advanced thermal protection fabrics. Spacesuits have multiple thermal protection layers made from various insulating fabrics to shield the wearer from extreme temperatures. Several candidate materials are being considered for their effectiveness in extreme temperature conditions. Twaron can endure up to 500

°C [35]. Twaron is a heat-resistant para-aramid fiber that is used for NASA's EVA gloves [27]. One major downside of Twaron is its relatively high absorption rate (2-7 %) [36]. In space, moisture can sublime, causing damage to spacesuits. Vectran can endure up to 400-450 °C [37], [38]. Vectran is a variant of Kevlar, and it is highly resistant to moisture, chemical damage, and tear but requires UV degradation protection [27]. It is one of the most promising materials overall.

6.1.1 Protection from Cosmic Radiation:

The space environment is inhospitable for humans to survive. Among the various impactful elements, cosmic radiation is one of the most dangerous elements in space. Cosmic radiation is composed of Galactic Cosmic Rays (GCR) - generally, originated in outer space, and Solar Cosmic Rays that are emitted by the Sun. Radiation protection is one of the key elements of the spacesuit design. The current spacesuit design uses a multilayer structure of various materials to protect astronauts from solar radiation and small space debris. Out of these layers, a layer of Kevlar and Nomux is used as protection against space debris (micrometeoroids in-specific) and hypothermia, and a layer of Mylar and aluminized film as protection against solar radiation. These multi-layers make the suit rigid and heavy.

A layer composed of hydrogen-rich compounds with lightweight atoms such as carbon and boron-10 could be used to attenuate the penetration of GCR and solar particles. Due to the composition of lightweight atoms, these materials weigh less, which contributes to making the spacesuit lightweight. Materials like RX1 and Z-graded (composition of atoms with different molecular weights) are under research. Based on the results made available publicly, it is understood that these materials showed better results compared to aluminized film even in attenuating the energy of secondary radiation. Per the materials tested to today's date, no material provides 100% protection from cosmic radiation by absorbing radiation entirely and not producing high-energy secondary radiation; the structure of multilayer of these materials can only interact with cosmic rays by deflecting or scattering them multiple times to attenuate the energy of radiation that penetrates through the protection layer.

6.1.2 Fire Resistant:

Fire resistance is a critical requirement for spacesuit materials due to the risk of fire hazards in the oxygen-rich environments of spacecraft and during extravehicular activities (EVAs). Traditional materials like Nomex and Kevlar are widely used for their inherent flame-resistant properties. Nomex, an aramid polymer, is particularly effective as it does not melt or drip when exposed to high temperatures, providing a crucial line of defense against fires. Kevlar, also an aramid fiber, offers high strength and durability in addition to its flame-resistant qualities, making it suitable for use in multiple layers of the spacesuit to provide comprehensive protection [39].

Recent advancements in materials science have introduced new fire-resistant coatings and treatments that can be applied to spacesuit fabrics to enhance their performance. For example, studies have shown that certain ceramic-based coatings can increase the thermal resistance of fabrics like Kevlar, providing additional protection against high-temperature flames.

The integration of these advanced materials and coatings is vital to ensure that spacesuits can withstand potential fire hazards, especially during critical phases of a mission, such as launch, re-entry, or EVAs [40].

6.1.3 Waterproof:

While Vectran is moisture-resistant, it is not inherently waterproof. Therefore, in many cases, Vectran is coated or combined with other materials to make it waterproof. One potential material of choice is polyurethane coating, which was previously presented by the Russian firm Zvezda NPP for use in ISS in 2015 [41]. Another material suitable for space missions is Polytetrafluoroethylene (PTFE). Similar to polyurethane coating, PTFE coating is also water resistant but provides higher UV light protection, compensating for Vectran's vulnerability to UV light when used together [42].

6.1.4 Lightweight:

Both Kevlar and PTFE coatings are widely used for their high resistance to multiple factors such as temperature, water, and radiation. In addition, they are extremely lightweight. Nomex is another alternative due to its lightweight property, but it is not as durable as materials like Kevlar. Furthermore, it is more vulnerable to moisture, potentially becoming heavier in humid environments.

6.2 Thermal Regulation:

Astronauts commonly wear the Liquid Cooling and Ventilation Garment (LCVG) to maintain a comfortable temperature by removing body heat. LCVG operates by circulating liquid around the wearer and cooling it again once it circulates back. LCVG is composed of three main parts: garments, tubes, and a heat exchanger. The garment, which is in direct contact with the wearer, requires both resistance and elasticity. Thus, the garment is often composed of materials such as Nylon Spandex. The tube component of LCVG is where the cooling liquid regulates body heat. It is made of silicone or PVC to withstand high pressure while providing high flexibility. The heat exchanger component of LCVG is a container used to cool the liquid again. While LCVG is highly effective at controlling the astronaut's body temperature, localized temperature control would offer additional benefits. Different parts of the body may produce different amounts of heat depending on the physical tasks being performed. Sensors monitoring body temperature and moisture can assist in providing optimal temperature for each part of the body. However, this may require a more complex tubing system.

6.3 Air Pressure in Spacesuit:

Maintaining proper air pressure within a spacesuit is essential for ensuring astronaut safety and comfort during EVAs. The pressure inside the suit must be carefully regulated to provide the necessary oxygen for breathing while preventing the risk of decompression sickness, also known as "the bends," which can occur if the pressure changes too rapidly.

Modern space suit designs utilize a combination of mechanical counterpressure (MCP) and gas pressurization to achieve the desired internal pressure. MCP suits, such as those utilizing elastic materials like spandex, apply pressure directly to the skin, reducing the need for higher gas pressures. This approach not only makes the suit more flexible and easier to move in but also decreases the pre-breathe time required before an EVA, thus improving operational efficiency.

Research is ongoing to improve the balance between suit flexibility and pressure regulation. Innovations in suit materials, such as the use of shape-memory alloys and advanced polymers, aim to maintain consistent pressure while allowing for greater mobility. Additionally, the development of smart textiles that can dynamically adjust pressure in response to

environmental conditions or the astronaut's movements is an area of active investigation.

These advancements are crucial for ensuring that spacesuits provide adequate protection while allowing astronauts to perform complex tasks in the challenging environments of space.

7. Applications

7.1 ISS Missions

For missions on the International Space Station (ISS), spacesuits must provide protection in a microgravity environment while ensuring ease of movement and durability. The new generation of spacesuits, such as those being developed by Collins Aerospace, emphasize flexibility and enhanced life support systems to improve astronaut performance during spacewalks. These suits incorporate advanced materials that offer better thermal regulation and micrometeoroid protection, essential for long-duration missions in low Earth orbit [43].



Figure 8. Prototype of Collins Aerospace Next-Generation Spacesuit [43]

7.2 Lunar Gateway and Lunar Missions

Lunar missions, including NASA's Artemis program and the Lunar Gateway, require spacesuits that are not only robust but also highly functional in the Moon's harsh environment. The AxEMU suit, developed by Axiom Space, is specifically designed to meet these demands. This suit offers superior mobility, crucial for both surface exploration and operations in the Lunar Gateway. It incorporates advanced materials to protect astronauts from extreme temperatures, radiation, and abrasive lunar dust, ensuring safety during extended missions [33], [44].

The AxEMU suit is an evolution of previous designs, integrating cutting-edge technology to support the varied and demanding applications of lunar exploration. Whether astronauts are conducting geological surveys on the lunar surface or maintaining systems in the Lunar Gateway, the suit provides the necessary protection and flexibility. This adaptability is essential for the success of both short-term missions and long-duration stays on or around the Moon.

7.3 Martian Missions

As humanity prepares for the future exploration of Mars, ensuring the durability and effectiveness of spacesuit materials is crucial. NASA is actively testing a variety of materials aboard the Perseverance rover to assess their ability to withstand Mars' harsh environment. This research focuses on evaluating the materials' resistance to the planet's extreme cold, intense radiation, and pervasive dust storms. The results from these tests will be instrumental in designing spacesuits that can protect astronauts and support extended missions on the Martian surface [45].

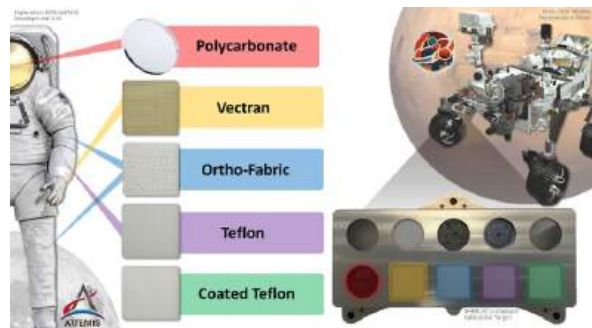


Figure 9. Spacesuit Material Samples on SHERLOC Calibration Target for Perseverance Rover [45]

7.4 Launch and Re-entry

Spacesuits used during launch and re-entry must offer strong protection against the extreme conditions encountered during these phases, including high G-forces, intense heat, and potential fire hazards. The Launch-Entry Suit (LES), used in the Space Shuttle program, incorporated fire-resistant materials like Nomex and Kevlar to safeguard astronauts. Similarly, SpaceX's Dragon launch and entry suits build on this legacy, offering advanced materials and design features that improve mobility, comfort, and safety during critical mission phases, ensuring astronaut protection in high-stress environments [46].

Conclusion

Space missions encompass various types of missions such as on-board ISS missions, spacewalks, planetary exploration, on-board deep space missions such as on-board Lunar Space Station (Gateway) missions, etc. As ISS is located in LEO, Earth's atmosphere plays a crucial role in protecting astronauts from cosmic radiation, therefore, additional protection beyond the spacecraft's radiation shield was not identified as a safety requirement. Therefore, currently, no suit is designed for the on-board ISS missions. However, in the future, on-board deep space missions will be conducted on the Lunar Space Station (Gateway) where spacecraft radiation shielding will not be sufficient to protect astronauts from the hazardous space environment. A suit needs to be designed for such on-board deep space mission applications.

Currently, a modified Apollo spacesuit is being used for spacewalks, which are massive, bulky, and rigid. For future planetary exploration missions, a lightweight, skinny, flexible, reusable, and low-maintenance suit is required. Several research programs are exploring techniques to design a single suit for spacewalks and planetary explorations. Japanese research programs are taking one step ahead in designing a suit that could be used for spacewalks, planetary explorations, and during launch and re-entry events. During the literature review, information on a suit design for on-board deep space missions was not found. Therefore, in the future, this paper will extend research on the suit design for on-board deep space missions.

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